Final Project Report: ZentroMart VoiceOps 2026

1. Project Overview

The ZentroMart VoiceOps project implements an automated customer support agent designed to handle diverse customer inquiries via chat and voice. The primary objective was to create a system that provides factually grounded answers based on a fictional company's knowledge base (KB), handles multi-turn conversations, performs secure identity verification, and manages escalations through ticket creation.

Scope of Solution:

- **Multi-modal Interface**: Integration of Gnani STT/TTS for voice interactions.
- **Grounded RAG**: A retrieval-augmented generation pipeline to ensure answers are derived from policy documents.
- **Deterministic Logic**: Hard-coded paths for high-certainty calculations (refund dates, return windows).
- **Secure Workflows**: Identity-checked order lookups and a structured escalation process.

2. System Architecture

2.1 Retrieval-Augmented Generation (RAG)

The system utilizes a local NumPy-based vector index built using the intfloat/multilingual-e5-small model.

- **Retrieval**: Uses a pool-16 relevance-aware ranking system to select the top-k most relevant knowledge chunks.
- **Grounding & Citations**: Every factual claim must be accompanied by a citation (e.g., [RET-2]). The system performs a structural check to ensure every cited ID was actually retrieved.
- **Response Parsing**: A robust JSON parser strips Markdown fences and validates the response schema, preventing "hallucinated" prose.

2.2 Deterministic Safeguards

To overcome LLM abstention bias and ensure 100% accuracy for critical data, deterministic paths were implemented:

- **Return Windows**: Logic that links a product's return_category directly to the RET-2 policy.
- **Serviceability**: Direct lookup of delivery status from pincodes.csv.
- **Dated Refunds**: A business-day calculator that accounts for Sundays and holidays.csv entries.

2.3 Conversation & Voice Architecture

- **Conversation Runner**: A stateful engine that tracks verification status, topic context, and ticket state.
- **Identity Verification**: Implements a two-attempt proof system (phone suffix or pincode) before releasing private order details.
- **Voice Pipeline**:
  - **Local Audio**: Successful validation of local microphone recording and speaker playback.
  - **STT**: Integration with Gnani STT was verified as functional in earlier testing.
  - **TTS**: Gnani SSE adapter synthesizing speech using the timbre-v2.5 model.
  - **Batch Processing**: A script to process folders of audio files into transcripts and WAV replies.

3. Evaluation and Results

3.1 RAG Performance

Evaluation against the 52 practice questions shows significant improvement over the LLM-only baseline:

Metric	Baseline (LLM-only)	RAG (v5)	Improvement
Non-abstention Rate	26.9%	84.6%	+57.7%
Abstention Rate	73.1%	15.4%	-57.7%
Citation Integrity	N/A	84.6%	New Capability
Parser Error Rate	N/A	0%	Resolved

Note: Non-abstention rate indicates the system provided a response rather than refusing due to lack of policy access; it does not inherently guarantee factual correctness. Citation integrity

confirms that cited evidence IDs were successfully retrieved, not their semantic accuracy.

- **Structural Stability**: The parser fix reduced invalid_structured_response cases from **31 (v4)** to **0 (v5)**.
- **Grounding**: Most responses now include required policy citations, with deterministic paths eliminating abstentions for key product-policy lookups (e.g., PQ0024, PQ0037).
- **Regression**: The offline test suite passed **47/47** tests, verifying the integrity of the retriever and RAG logic.

3.2 Retrieval Metrics

The retriever demonstrates strong performance in surfacing necessary policy sections, achieving an average **Recall@K of 91%** and an **MRR (Mean Reciprocal Rank) of 0.64** across the practice dataset.

4. Limitations and Future Work

4.1 Identified Blockers

- **Gnani TTS Connectivity**: Live TTS requests consistently returned **HTTP 403 Forbidden (Error 1010)**. While the adapter logic and local audio playback are verified, the cause of this API rejection was not conclusively established.
- **End-to-End Voice Validation**: Because the TTS component remained blocked, the complete end-to-end voice-to-voice workflow was not fully validated.

4.2 Future Improvements

- **Semantic Validation**: Moving beyond structural citation checks to automated semantic verification.
- **Expanded Tooling**: Integrating real-time ticket tracking and live agent handoff.

5. Reproducibility

5.1 Environment Setup

```
powershell python -m venv .venv .\.venv\Scripts\activate pip install -r requirements.txt
```

5.2 Execution Commands

- **Index Build**: python build_rag_index.py
- **RAG Evaluation**: python run_rag.py --output outputs/rag_results.csv
- **Offline Tests**: .venv\Scripts\python.exe -m pytest tests/test_rag.py
- **Batch Voice**: python scripts/run_voice_batch.py --input-dir .\audio\incoming --output-dir .\outputs\voice_batch

---

Date: 25 October 2026 **Project**: ZentroMart VoiceOps Capstone